Modeling Periodic Behavior from Graphs
Answer Key
Precalculus

Quick Answers

1. (a) amplitude = 3, (b) midline value = 2
2. (a) amplitude = 2, (b) midline value = 1
3. value of B = $\frac{1}{2}$
4. (a) amplitude = 5, (b) midline value = −3
5. value of B = 3
6. (a) A = 4, (b) D = 5, (c) B = 2
7. times when the depth is 8 metres = 1, 5 hours
8. value of B = $\frac{\pi}{4}$
9. (a) amplitude A, in metres = 20 m, (b) B = $\frac{\pi}{4}$, —
10. times when the temperature is 24 degrees = 2, 10 hours
11. (a) amplitude = 3, (b) midline value = 1.5, —
12. (a) A, in metres = 22 m, (b) D, in metres = 24 m, (c) B = $\frac{\pi}{4}$, (d) times at a height of 35 metres = $\frac{8}{3}$, $\frac{16}{3}$ minutes

What is verified

22 of 24 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 9, 11. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSF-TF.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 24 tagged checks.

1. Graph of $y = 3\sin x + 2$: maximum 5, minimum −1. (a) Amplitude. (b) Midline value.

(a) Amplitude is half the vertical spread between the extremes: $\dfrac{5-(-1)}{2} = \dfrac{6}{2} = 3$.

$\boxed{A = 3}$

(b) The midline sits halfway between them: $\dfrac{5+(-1)}{2} = \dfrac{4}{2} = 2$, so the midline is $y = 2$.

$\boxed{D = 2}$

2. Graph of $y = -2\cos x + 1$: maximum 3, minimum −1. (a) Amplitude. (b) Midline value.

(a) $\dfrac{3-(-1)}{2} = 2$. Amplitude is a distance, so it is positive even though this curve opens downward first.

$\boxed{A = 2}$

(b) $\dfrac{3+(-1)}{2} = 1$, so the midline is $y = 1$. $\boxed{D = 1}$

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3. One cycle of $y = A\cos(Bx)$ runs from $x = 0$ to $x = 4\pi$. Find B.

Solution: the graph completes one cycle over 4π, so the period is 4π. For a sinusoid the period is $\dfrac{2\pi}{B}$, so

$$B = \frac{2\pi}{\text{period}} = \frac{2\pi}{4\pi} = \frac{1}{2}.$$

A period longer than 2π always gives B between 0 and 1, which is a useful check. $\boxed{B = \frac{1}{2}}$

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4. Graph of $y = 5\sin x - 3$: maximum 2, minimum -8. (a) Amplitude. (b) Midline value.

(a) $\dfrac{2-(-8)}{2} = \dfrac{10}{2} = 5$. $\boxed{A = 5}$

(b) $\dfrac{2+(-8)}{2} = \dfrac{-6}{2} = -3$, so the midline is $y = -3$ — below the x-axis, which is exactly why amplitude cannot be read as the maximum. $\boxed{D = -3}$

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5. Graph of $y = \sin(Bx)$: one cycle runs from $x = 0$ to $x = \dfrac{2\pi}{3}$. Find B.

Solution: the curve returns to the start of its pattern at $x = \dfrac{2\pi}{3}$, so that is the period:

$$B = \frac{2\pi}{2\pi/3} = 2\pi \cdot \frac{3}{2\pi} = 3.$$

Three complete cycles fit in 2π, which matches the picture. $\boxed{B = 3}$

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6. Graph modelled by $y = A\cos(Bx) + D$: maximum 9, minimum 1, period π. (a) A. (b) D. (c) B.

(a) $A = \dfrac{9-1}{2} = 4$. $\boxed{A = 4}$

(b) $D = \dfrac{9+1}{2} = 5$. $\boxed{D = 5}$

(c) The maximum recurs every π units, so the period is π and $B = \dfrac{2\pi}{\pi} = 2$. The model is $y = 4\cos(2x) + 5$; because the graph peaks at $x = 0$, cosine needs no shift. $\boxed{B = 2}$

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7. $d(t) = 2\sin\left(\dfrac{\pi t}{6}\right) + 7$ metres. Find all t in $0 \le t < 12$ with $d = 8$.

Solution: isolate the sine, then use the two angles in $[0, 2\pi)$ with that sine.

$$\begin{aligned} 2\sin\left(\frac{\pi t}{6}\right) + 7 &= 8 \\ \sin\left(\frac{\pi t}{6}\right) &= \frac{1}{2} \\ \frac{\pi t}{6} = \frac{\pi}{6} \quad &\text{or} \quad \frac{\pi t}{6} = \frac{5\pi}{6} \end{aligned}$$

Multiplying by $\frac{6}{\pi}$ gives $t = 1$ and $t = 5$. As t runs over $[0, 12)$ the angle $\frac{\pi t}{6}$ runs over $[0, 2\pi)$ exactly once, so there are no further solutions. $\boxed{t = 1,\ 5 \text{ hours}}$

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8. Two complete cycles fill 16 seconds on the sensor graph. For $y = A\sin(Bt)$, find B.
Solution: two cycles in 16 seconds means one cycle in 8 seconds, so the period is 8:

$$B = \frac{2\pi}{8} = \frac{\pi}{4}.$$

$\boxed{B = \frac{\pi}{4}}$

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9. Wheel of diameter 40 m, centre 22 m up, one turn every 8 minutes, $h(t) = A\sin(Bt) + 22$. (a) A. (b) B. (c) Sketch two revolutions.

(a) The rider moves from the centre out to the rim, so the amplitude is the radius — half the diameter: $\frac{40}{2} = 20$. $\boxed{A = 20 \text{ m}}$

(b) One revolution is one period, so the period is 8 minutes and $B = \frac{2\pi}{8} = \frac{\pi}{4}$. The model is $h(t) = 20\sin\left(\frac{\pi t}{4}\right) + 22$. $\boxed{B = \frac{\pi}{4}}$

(c) *Instructor-judged.* The sketch must show two complete revolutions across the 16 minute grid, starting at the centre height 22 and rising first. Check for: a horizontal midline at 22; peaks at 42 and troughs at 2; the first peak at $t = 2$ and the first trough at $t = 6$; a smooth curve, not straight segments. Labels on the midline, the maximum and the minimum are required for full credit.

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10. $T(t) = 8\sin\left(\frac{\pi t}{12}\right) + 20$ degrees. Find all t in $0 \le t < 24$ with $T = 24$.

Solution:

$$\begin{aligned} 8\sin\left(\frac{\pi t}{12}\right) + 20 &= 24 \\ \sin\left(\frac{\pi t}{12}\right) &= \frac{1}{2} \\ \frac{\pi t}{12} = \frac{\pi}{6} \quad &\text{or} \quad \frac{\pi t}{12} = \frac{5\pi}{6} \end{aligned}$$

Multiplying by $\frac{12}{\pi}$: $t = 2$ and $t = 10$. Over $[0, 24)$ the angle covers $[0, 2\pi)$ once, so the list is complete — and the graph confirms the curve crosses $T = 24$ twice, once rising and once falling. $\boxed{t = 2,\ 10 \text{ hours}}$

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11. Graph with maximum 4.5 and minimum -1.5. (a) Amplitude. (b) Midline value. (c) Explain why the amplitude is not the maximum.

(a) $\dfrac{4.5 - (-1.5)}{2} = \dfrac{6}{2} = 3.$ $\boxed{A = 3}$

(b) $\dfrac{4.5 + (-1.5)}{2} = \dfrac{3}{2} = 1.5$, so the midline is $y = 1.5$. $\boxed{D = 1.5}$

(c) *Instructor-judged.* A correct response says the maximum is measured from the horizontal axis while the amplitude is measured from the *midline*, and the two agree only when the midline happens to lie on the axis. Here the midline is $y = 1.5$, so the amplitude is $4.5 - 1.5 = 3$, not 4.5. Repeating the arithmetic without naming the midline as the reference level is partial credit.

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12. Wheel graph: maximum 46 m, minimum 2 m, first maximum at $t = 4$ min, period 8 min, $h(t) = A\cos\big(B(t-4)\big) + D$. (a) A. (b) D. (c) B. (d) Times at height 35 m.

(a) $A = \dfrac{46 - 2}{2} = 22.$ $\boxed{A = 22 \text{ m}}$

(b) $D = \dfrac{46 + 2}{2} = 24.$ $\boxed{D = 24 \text{ m}}$

(c) One revolution is the period, 8 minutes, so $B = \dfrac{2\pi}{8} = \dfrac{\pi}{4}$. The model is $h(t) = 22\cos\Big(\dfrac{\pi}{4}(t-4)\Big) + 24$; the shift of 4 puts cosine's peak where the graph peaks. $\boxed{B = \frac{\pi}{4}}$

(d) Set $h = 35$ and isolate the cosine:

$$22\cos\left(\frac{\pi(t-4)}{4}\right) + 24 = 35$$
$$\cos\left(\frac{\pi(t-4)}{4}\right) = \frac{11}{22} = \frac{1}{2}$$
$$\frac{\pi(t-4)}{4} = \frac{\pi}{3} \quad \text{or} \quad \frac{\pi(t-4)}{4} = -\frac{\pi}{3}$$

Multiplying by $\frac{4}{\pi}$ gives $t - 4 = \pm\frac{4}{3}$, so $t = 4 - \frac{4}{3} = \frac{8}{3}$ and $t = 4 + \frac{4}{3} = \frac{16}{3}$. Both lie in $[0, 8)$, and the symmetry about the peak at $t = 4$ is the check. $\boxed{t = \frac{8}{3},\ \frac{16}{3} \text{ minutes}}$